1 Making Decisions under Uncertainty

2 Convergent Prior

Definition 2.1 (probabilistic model)**.** We define a *probabilistic model* as a pair $(\mathcal{S}, \mathcal{P})$, where $\mathcal{S}$ is a sample space, and $\mathcal{P}$ is a set of *acceptable* probability measures on $\mathcal{S}$.

Definition 2.2 (ϵ-convergence)**.** Let S and E be random variables in a probabilistic model $\mathcal{M}$. We say $S|E$ (S given E) *ϵ-converges* by a distribution P in $\mathcal{M}$, if for every distribution Q in $\mathcal{M}$ we have:

$$\mathbb{E}_Q \left[\mathbb{D}(P_{S|E} || Q_{S|E})\right] = \sum_{s \in \mathcal{S}} \sum_{e \in \mathcal{E}} \ln \frac{Q(s \mid e)}{P(s \mid e)} Q(s, e) < \epsilon \cdot \tag{1}$$

Definition 2.3 (ϵ-convergent distribution)**.** Let S be a random variable, and $\mathbf{E}$ be a set of random variables in a probabilistic model $\mathcal{M}$. We say P is an *ϵ-convergent distribution* for $S|\mathbf{E}$ in $\mathcal{M}$, if for all $E \in \mathbf{E}$, $S|E$ ϵ-converges by P in $\mathcal{M}$.

Theorem 1. *Consider two probabilistic models $\mathcal{M}$ and $\mathcal{M}^*$. Let $\mathbf{E}$ be a* ***countable*** *set of random variables in $\mathcal{M}$, and E^* be a single random variable in $\mathcal{M}^*$. Suppose that there exists a random variable I in $\mathcal{M}^*$, such that for* ***every*** *probability measure P in $\mathcal{M}$, there exists a probability measure P^* in $\mathcal{M}^*$, such that for every $E_i \in \mathbf{E}$, every $s \in \mathcal{S}$ and every $e_i \in \mathcal{E}_i$, there exist some values s^*, e^* such that we have:*

$$P(S = s, E_i = e_i) = P^*(S^* = s^*, E^* = e^* \mid I = i) \cdot \tag{2}$$

In addition, I and E^* are conditionally independent:

$$P^*(S^* \mid E^*) = P^*(S^* \mid E^*, I) \cdot \tag{3}$$

Then $S|\mathbf{E}$ ϵ-converges by P in $\mathcal{M}$, if $S^*|E^*$ ϵ-converges by P^* in $\mathcal{M}^*$.

Proof. Assume, by contradiction, that $S|\mathbf{E}$ does not ϵ-converge by P in $\mathcal{M}$. Therefore, there must exist Q and $E_i \in \mathbf{E}$ in $\mathcal{M}$ such that:

$$\mathbb{E}_Q \left[\mathbb{D}(P_{S|E_i} || Q_{S|E_i})\right] \geq \epsilon \cdot$$

By using equations 2 and 3 we can rewrite the left hand side:

$$\begin{aligned}\mathbb{E}_Q\left[\mathbb{D}(P_{S|E_i}||Q_{S|E_i})\right] &= \sum_{s\in\mathcal{S}}\sum_{e_i\in\mathcal{E}_i}\ln\frac{Q(s\mid e_i)}{P(s\mid e_i)}Q(s,e_i)\\ &= \sum_{s^*\in\mathcal{S}^*}\sum_{e^*\in\mathcal{E}^*}\ln\frac{Q^*(s^*\mid e^*, I=i)}{P^*(s^*\mid e^*, I=i)}Q^*(s^*,e^*\mid I=i)\\ &= \sum_{s^*\in\mathcal{S}^*}\sum_{e^*\in\mathcal{E}^*}\ln\frac{Q^*(s^*\mid e^*)}{P^*(s^*\mid e^*)}Q^*(s^*,e^*\mid I=i)\ \cdot\end{aligned}$$

Hence, we have:

$$\sum_{s^*\in\mathcal{S}^*}\sum_{e^*\in\mathcal{E}^*}\ln\frac{Q^*(s^*\mid e^*)}{P^*(s^*\mid e^*)}Q^*(s^*,e^*\mid I=i)\geq\epsilon\ \cdot$$

Since $Q^*(i)$ is a distribution and is non-negative for every i, we can multiply both sides of this inequality to $Q^*(i)$ and sum over all values of i:

$$\begin{gathered}\sum_i\sum_{s^*\in\mathcal{S}^*}\sum_{e^*\in\mathcal{E}^*}\ln\frac{Q^*(s^*\mid e^*)}{P^*(s^*\mid e^*)}Q^*(s^*,e^*,I=i)\geq\sum_i\epsilon Q^*(i)\\ \sum_{s^*\in\mathcal{S}^*}\sum_{e^*\in\mathcal{E}^*}\ln\frac{Q^*(s^*\mid e^*)}{P^*(s^*\mid e^*)}\sum_i Q^*(s^*,e^*,I=i)\geq\epsilon\sum_i Q^*(i)\\ \sum_{s^*\in\mathcal{S}^*}\sum_{e^*\in\mathcal{E}^*}\ln\frac{Q^*(s^*\mid e^*)}{P^*(s^*\mid e^*)}Q^*(s^*,e^*)\geq\epsilon\ ,\end{gathered}$$

which means $S^*|E^*$ does not ϵ-converge by P^* in $\mathcal{M}^*$. □

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References

[1] Nobody Jr. My article, 2006.